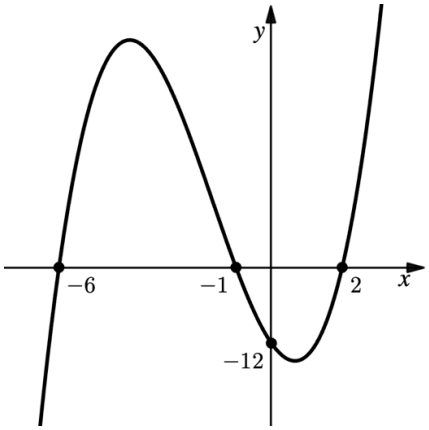
1.16 Inequalities: Polynomials

Worked Solutions

1 a)



1 b)

i)

$$\{x : x \in \mathbb{R}, -6 < x < -1\} \cup \{x : x \in \mathbb{R}, x > 2\}$$

ii)

$$\{x : x \in \mathbb{R}, x \leq -6\} \cup \{x : x \in \mathbb{R}, -1 \leq x \leq 2\}$$

1 c)

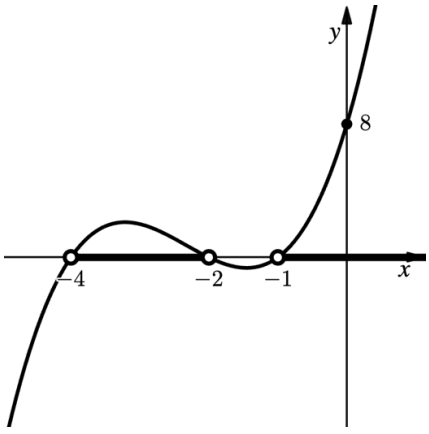
i)

$$x \in (-\infty, -6) \cup (-1, 2)$$

ii)

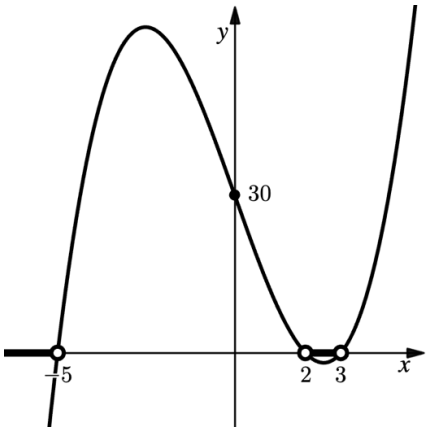
$$x \in [-6, -1] \cup [2, \infty)$$

2 a)



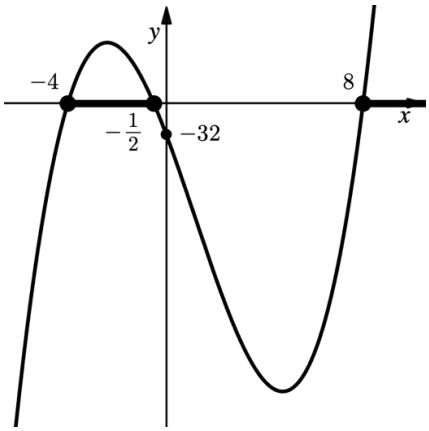
$$\{x : x \in \mathbb{R}, -4 < x < -2\} \cup \{x : x \in \mathbb{R}, x > -1\}$$

2 b)



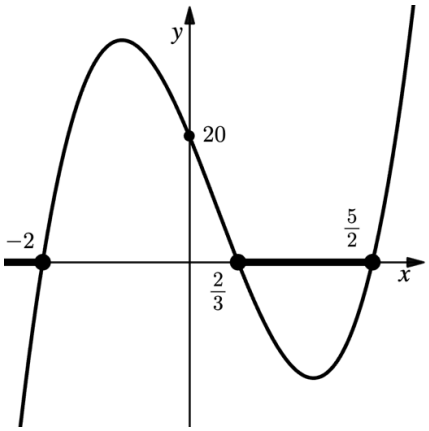
$$\{x : x \in \mathbb{R}, x < -5\} \cup \{x : x \in \mathbb{R}, 2 < x < 3\}$$

2 c)



$$\{x : x \in \mathbb{R}, -4 \leq x \leq -\frac{1}{2}\} \cup \{x : x \in \mathbb{R}, x \geq 8\}$$

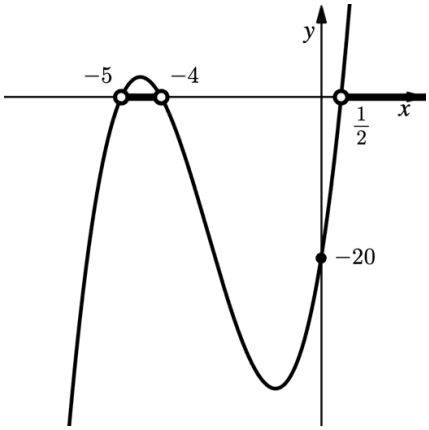
2 d)



$$\{x : x \in \mathbb{R}, x \leq -2\} \cup \{x : x \in \mathbb{R}, \frac{2}{3} \leq x \leq \frac{5}{2}\}$$

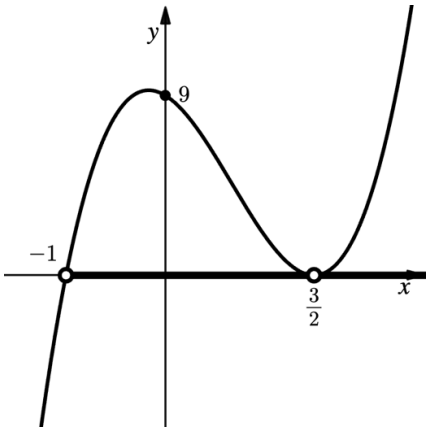
2 e)

$$(2x-1)(x^2+9x+20) > 0$$
$$\Rightarrow (2x-1)(x+4)(x+5) > 0$$



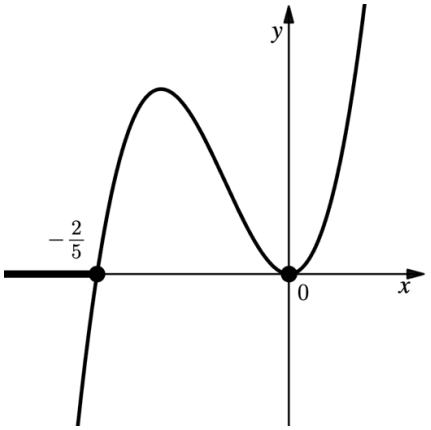
$$\{x : x \in \mathbb{R}, -5 < x < -4\} \cup \{x : x \in \mathbb{R}, x > \frac{1}{2}\}$$

2 f)



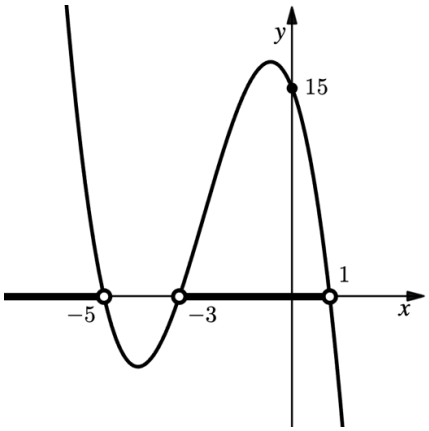
$$\{x : x \in \mathbb{R}, -1 < x < \frac{3}{2}\} \cup \{x : x \in \mathbb{R}, x > \frac{3}{2}\}$$

2 g)



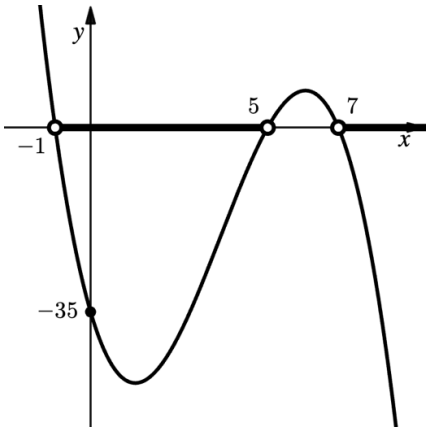
$$\{x : x \in \mathbb{R}, x \leq -\frac{2}{5}\} \cup \{x : x \in \mathbb{R}, x = 0\}$$

3 a)



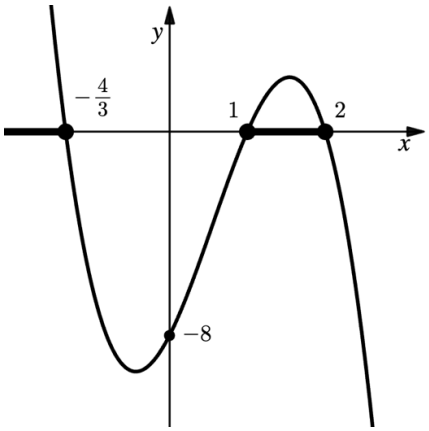
$$\{x : x \in \mathbb{R}, x < -5\} \cup \{x : x \in \mathbb{R}, -3 < x < 1\}$$

3 b)



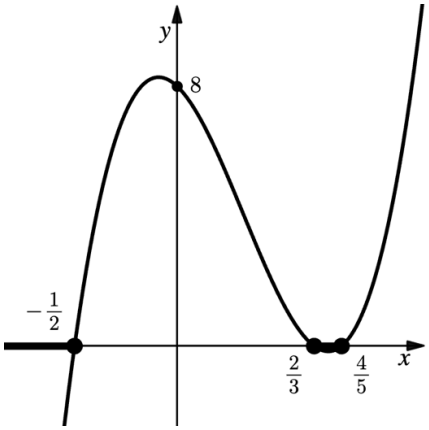
$$\{x : x \in \mathbb{R}, -1 < x < 5\} \cup \{x : x \in \mathbb{R}, x > 7\}$$

3 c)



$$\{x : x \in \mathbb{R}, x \leq -\frac{4}{3}\} \cup \{x : x \in \mathbb{R}, 1 \leq x \leq 2\}$$

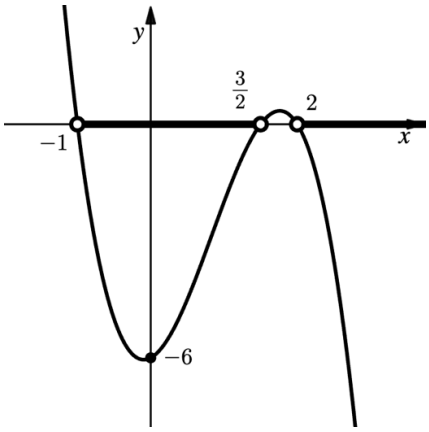
3 d)



$$\{x : x \in \mathbb{R}, x \leq -\frac{1}{2}\} \cup \{x : x \in \mathbb{R}, \frac{2}{3} \leq x \leq \frac{4}{5}\}$$

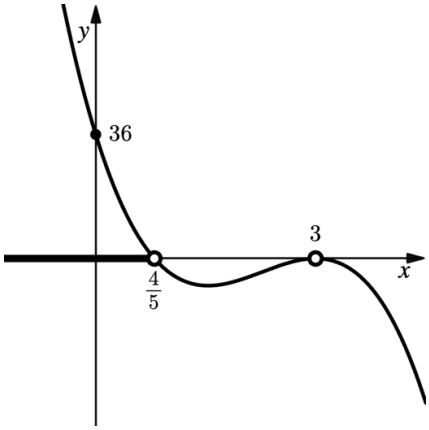
3 e)

$$(3-2x)(x^2-x-2)<0$$
$$\Rightarrow (3-2x)(x-2)(x+1)<0$$



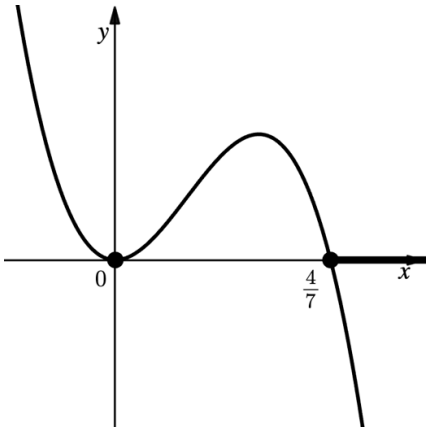
$$\{x : x \in \mathbb{R}, -1 < x < \frac{3}{2}\} \cup \{x : x \in \mathbb{R}, x > 2\}$$

3 f)



$$\{x : x \in \mathbb{R}, x < \frac{4}{5}\}$$

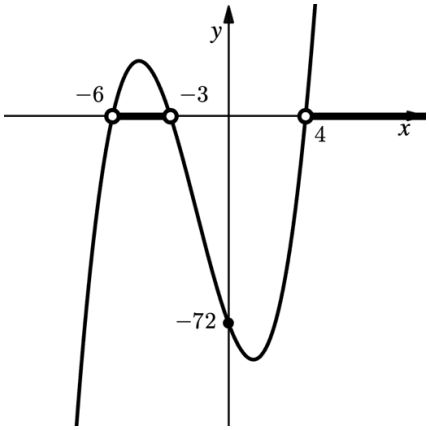
3 g)



$$\{x : x \in \mathbb{R}, x \geq \frac{4}{7}\}$$

4 a)

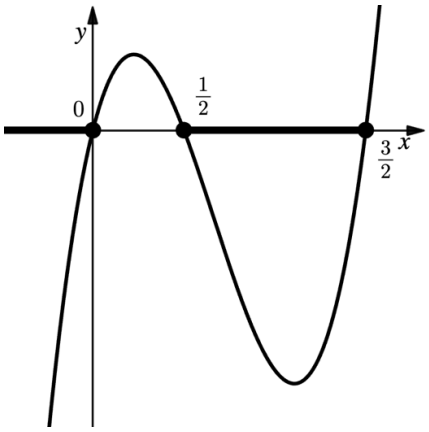
$$x^3 + 5x^2 - 18x - 72 > 0$$
$$\Rightarrow (x+3)(x-4)(x+6) > 0$$



$$x \in (-6, -3) \cup (4, \infty)$$

4 b)

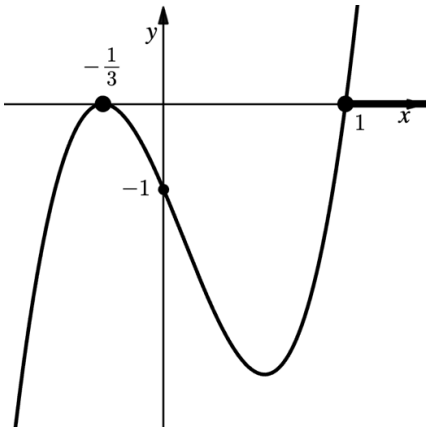
$$4x^3 - 8x^2 + 3x \leq 0$$
$$\Rightarrow (2x-1)(2x-3)x \leq 0$$



$$x \in (-\infty, 0] \cup [\frac{1}{2}, \frac{3}{2}]$$

4 c)

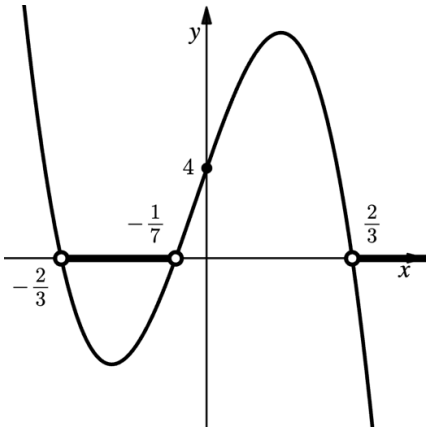
$$9x^3 - 3x^2 - 5x - 1 \geq 0$$
$$\Rightarrow (3x+1)^2(x-1) \geq 0$$



$$x \in \{-\frac{1}{3}\} \cup [1, \infty) \text{ or } x \in [-\frac{1}{3}, -\frac{1}{3}] \cup [1, \infty)$$

4 d)

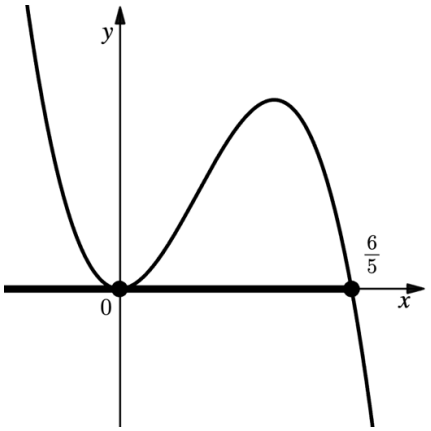
$$4 + 28x - 9x^2 - 63x^3 < 0$$
$$\Rightarrow (2 - 3x)(7x + 1)(3x + 2) < 0$$



$$x \in (-\frac{2}{3}, -\frac{1}{7}) \cup (\frac{2}{3}, \infty)$$

4 e)

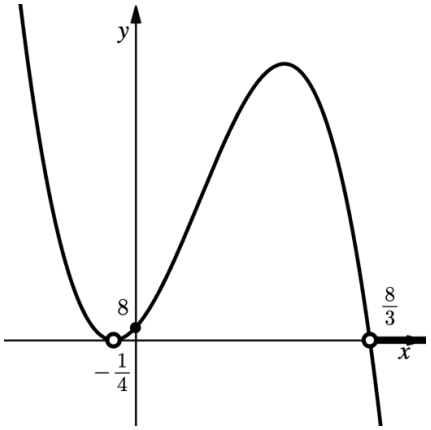
$$6x^2 - 5x^3 \geq 0$$
$$\Rightarrow (6 - 5x)x^2 \geq 0$$



$$x \in (-\infty, \frac{6}{5}]$$

4 f)

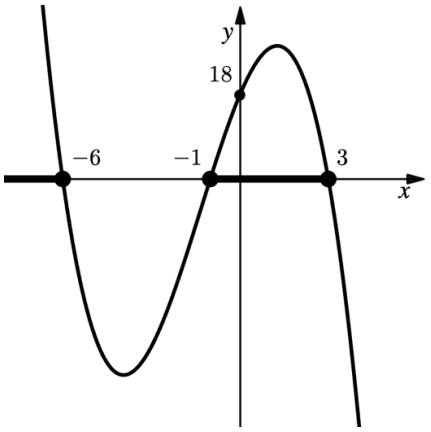
$$8 + 61x + 104x^2 - 48x^3 < 0$$
$$\Rightarrow (8 - 3x)(4x + 1)^2 < 0$$



$$x \in (\frac{8}{3}, \infty)$$

5 a)

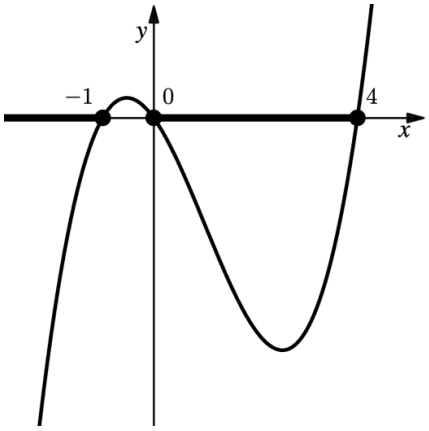
$$(1 + x)(3 - x)(x + 5) \geq (1 + x)(x - 3)$$
$$\Rightarrow (1 + x)(3 - x)(x + 5) - (1 + x)(x - 3) \geq 0$$
$$\Rightarrow (1 + x)(3 - x)(x + 5) + (1 + x)(3 - x) \geq 0$$
$$\Rightarrow (1 + x)(3 - x)((x + 5) + 1) \geq 0$$
$$\Rightarrow (1 + x)(3 - x)(x + 6) \geq 0$$



$$\{x : x \in \mathbb{R}, x \leq -6\} \cup \{x : x \in \mathbb{R}, -1 \leq x \leq 3\}$$

5 b)

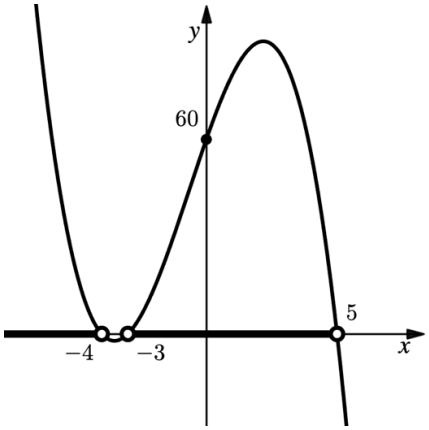
$$
\begin{aligned}
&(2+x)(x-4)x \leq (4-x)x^2 \\
\Rightarrow\ &(2+x)(x-4)x - (4-x)x^2 \leq 0 \\
\Rightarrow\ &(2+x)(x-4)x + (x-4)x^2 \leq 0 \\
\Rightarrow\ &x(x-4)((2+x)+x) \leq 0 \\
\Rightarrow\ &x(x-4)(2+2x) \leq 0 \\
\Rightarrow\ &x(x-4)(1+x) \leq 0
\end{aligned}
$$



$$\{x : x \in \mathbb{R}, x \leq -1\} \cup \{x : x \in \mathbb{R}, 0 \leq x \leq 4\}$$

5 c)

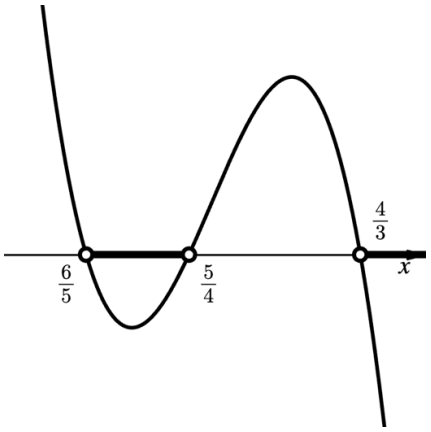
$$
\begin{aligned}
&(5-x)(3+x) > (3+x)^2(x-5) \\
\Rightarrow\ &(5-x)(3+x) - (3+x)^2(x-5) > 0 \\
\Rightarrow\ &(5-x)(3+x) + (3+x)^2(5-x) > 0 \\
\Rightarrow\ &(5-x)(3+x)(1+(3+x)) > 0 \\
\Rightarrow\ &(5-x)(3+x)(4+x) > 0
\end{aligned}
$$



$$\{x : x \in \mathbb{R}, x < -4\} \cup \{x : x \in \mathbb{R}, -3 < x < 5\}$$

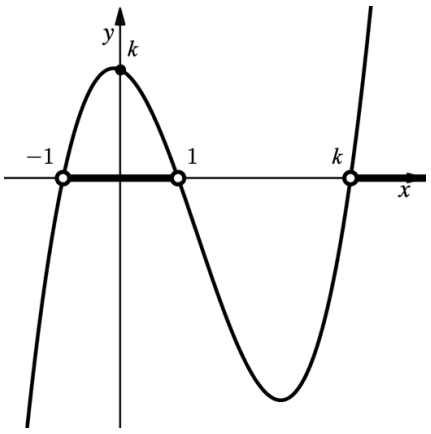
5 d)

$$(4-3x)^2(6-5x) < (3x-4)(5x-6)^2$$
$$\Rightarrow (3x-4)^2(6-5x) - (3x-4)(6-5x)^2 < 0$$
$$\Rightarrow (3x-4)(6-5x)((3x-4)-(6-5x)) < 0$$
$$\Rightarrow (3x-4)(6-5x)(8x-10) < 0$$
$$\Rightarrow (3x-4)(6-5x)(4x-5) < 0$$



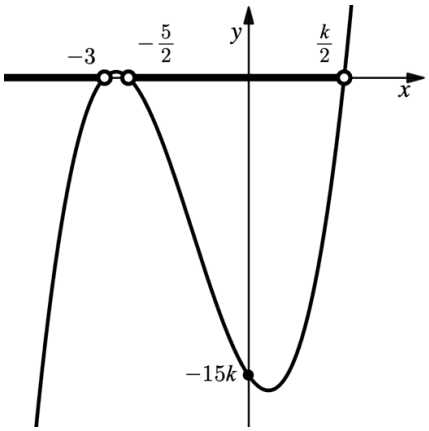
$$\{x : x \in \mathbb{R}, \frac{6}{5} < x < \frac{5}{4}\} \cup \{x : x \in \mathbb{R}, x > \frac{4}{3}\}$$

6 a)



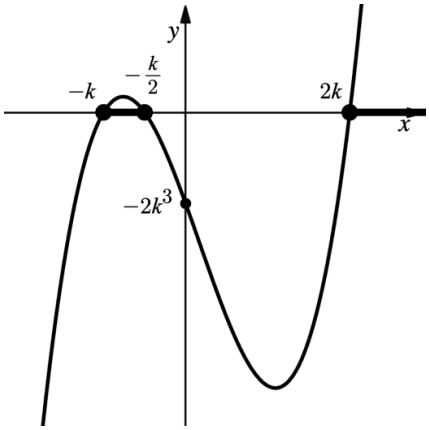
$$\{x : x \in \mathbb{R}, -1 < x < 1\} \cup \{x : x \in \mathbb{R}, x > k\}$$

6 b)



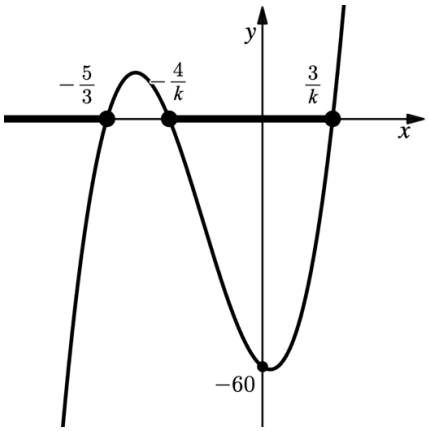
$$\{x : x \in \mathbb{R}, x < -3\} \cup \{x : x \in \mathbb{R}, -\frac{5}{2} < x < \frac{k}{2}\}$$

6 c)



$$\{x : x \in \mathbb{R}, -k \leq x \leq -\frac{k}{2}\} \cup \{x : x \in \mathbb{R}, x \geq 2k\}$$

6 d)



$$\{x : x \in \mathbb{R}, x \leq -\frac{5}{3}\} \cup \{x : x \in \mathbb{R}, -\frac{4}{k} \leq x \leq \frac{3}{k}\}$$

7)
Critical values are $x = -\frac{7}{2}$, $x = -2$ and $x = q$

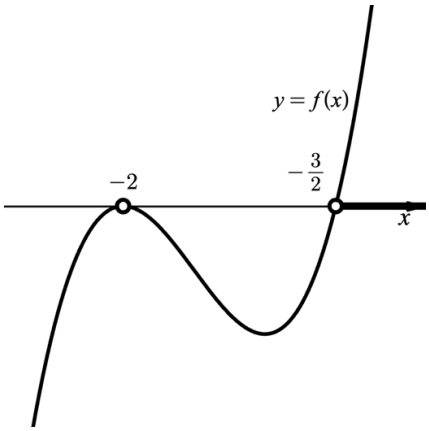
$$\begin{gathered}
\therefore (2x+7)(x+2)(x-q) \geq 0 \\
\Rightarrow (2x^2 - 2qx + 7x - 7q)(x+2) \geq 0 \\
\Rightarrow (2x^2 + (7-2q)x - 7q)(x+2) \geq 0 \\
7 - 2q = 19 \Rightarrow q = -6 \\
p = -7(-6) = 42
\end{gathered}$$

8)
Critical values are $x = -4$, $x = -\frac{9}{4}$ and $x = r$

$$\begin{gathered}
\therefore (x+4)(4x+9)(x-r) \leq 0 \\
\Rightarrow (x+4)(4x^2 - 4rx + 9x - 9r) \leq 0 \\
\Rightarrow (x+4)(4x^2 + (9-4r)x - 9r) \leq 0 \\
\Rightarrow 4x^3 + (9-4r)x^2 - 9rx + 16x^2 + 4(9-4r)x - 36r \leq 0 \\
\Rightarrow 4x^3 + (25-4r)x^2 + (36-25r)x - 36r \leq 0 \\
25 - 4r = 17 \Rightarrow r = 2 \\
q = 36 - 25(2) = -14
\end{gathered}$$

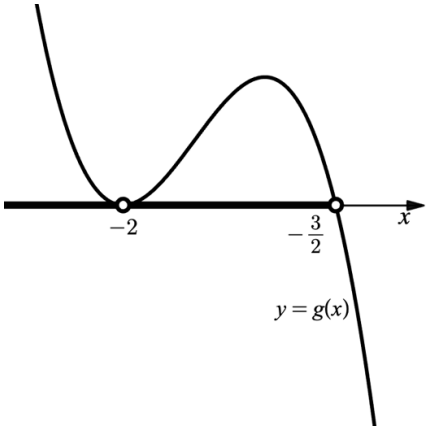
So $p = 4$, $q = -14$ and $r = 2$

9 a)



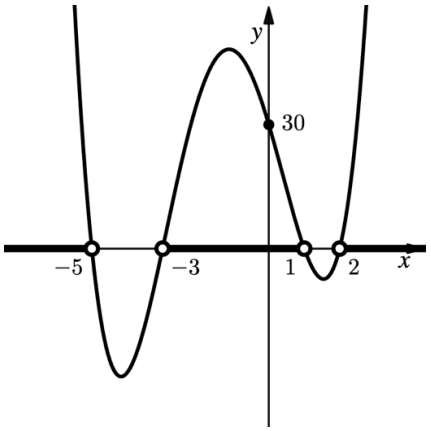
$$x \in (-\frac{3}{2}, \infty)$$

9 b)



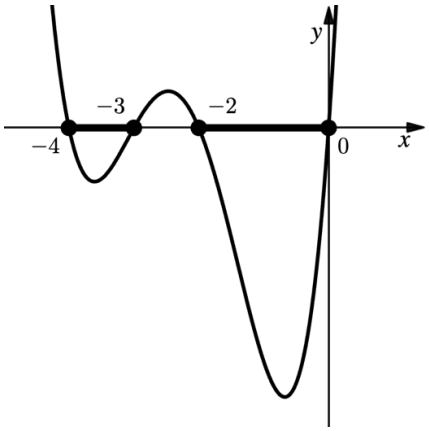
$$x \in (-\infty, -2) \cup (-2, -\frac{3}{2})$$

10 a)



$$\{x : x \in \mathbb{R}, x < -5\} \cup \{x : x \in \mathbb{R}, -3 < x < 1\} \cup \{x : x \in \mathbb{R}, x > 2\}$$

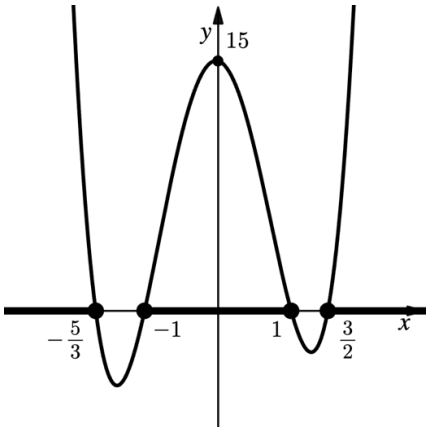
10 b)



$$\{x : x \in \mathbb{R}, -4 \leq x \leq -3\} \cup \{x : x \in \mathbb{R}, -2 \leq x \leq 0\}$$

10 c)

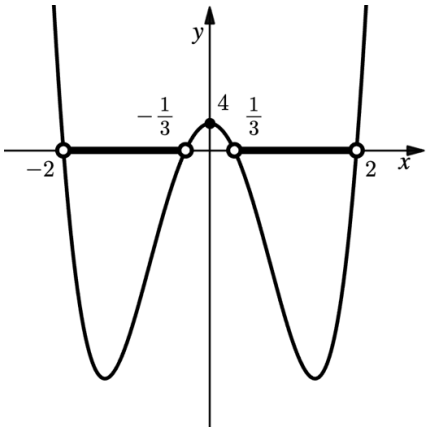
$$(2x-3)(3x+5)(x^2-1) \geq 0$$
$$\Rightarrow (2x-3)(3x+5)(x-1)(x+1) \geq 0$$



$$\{x : x \in \mathbb{R}, x \leq -\frac{5}{3}\} \cup \{x : x \in \mathbb{R}, -1 \leq x \leq 1\} \cup \{x : x \in \mathbb{R}, x \geq \frac{3}{2}\}$$

10 d)

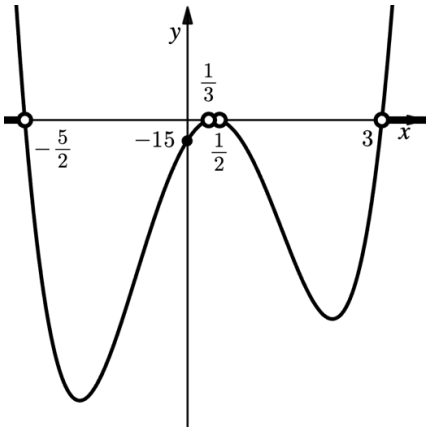
$$(x^2-4)(9x^2-1) < 0$$
$$\Rightarrow (x-2)(x+2)(3x-1)(3x+1) < 0$$



$$\{x : x \in \mathbb{R}, -2 < x < -\frac{1}{3}\} \cup \{x : x \in \mathbb{R}, \frac{1}{3} < x < 2\}$$

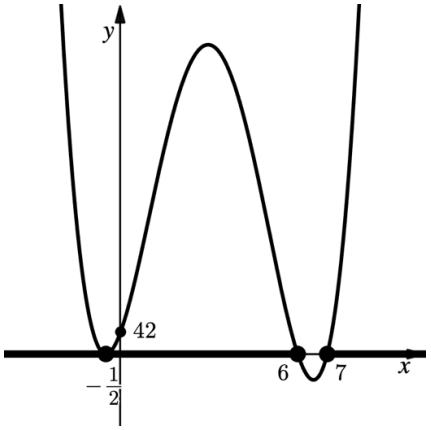
10 e)

$$(4x^2 + 8x - 5)(3x^2 - 10x + 3) > 0$$
$$\Rightarrow (2x - 1)(2x + 5)(3x - 1)(x - 3) > 0$$



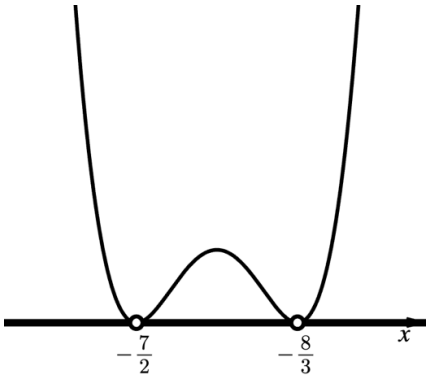
$$\{x : x \in \mathbb{R}, x < -\frac{5}{2}\} \cup \{x : x \in \mathbb{R}, \frac{1}{3} < x < \frac{1}{2}\} \cup \{x : x \in \mathbb{R}, x > 3\}$$

10 f)



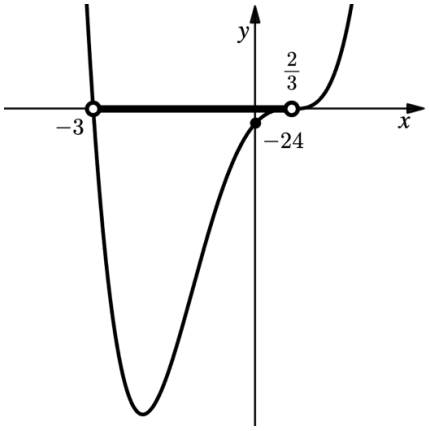
$$\{x : x \in \mathbb{R}, x \leq 6\} \cup \{x : x \in \mathbb{R}, x \geq 7\}$$

10 g)



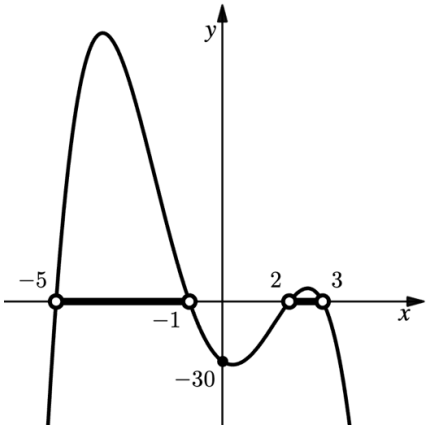
$$\{x : x \in \mathbb{R}, x < -\frac{7}{2}\} \cup \{x : x \in \mathbb{R}, -\frac{7}{2} < x < -\frac{8}{3}\} \cup \{x : x \in \mathbb{R}, x > -\frac{8}{3}\}$$

10 h)



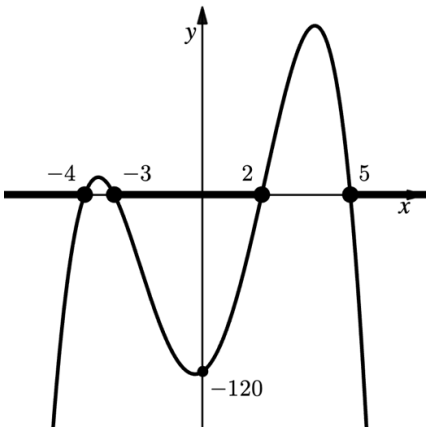
$$\{x : x \in \mathbb{R}, -3 < x < \frac{2}{3}\}$$

11 a)



$$\{x : x \in \mathbb{R}, -5 < x < -1\} \cup \{x : x \in \mathbb{R}, 2 < x < 3\}$$

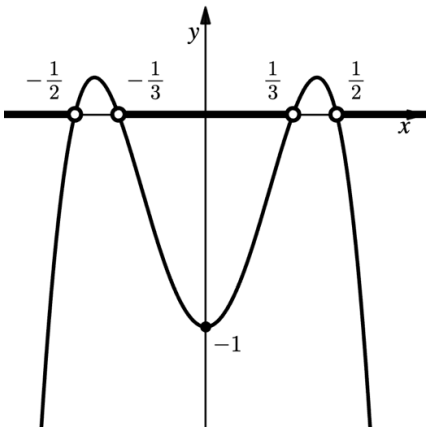
11 b)



$$\{x : x \in \mathbb{R}, x \leq -4\} \cup \{x : x \in \mathbb{R}, -3 \leq x \leq 2\} \cup \{x : x \in \mathbb{R}, x \geq 5\}$$

11 c)

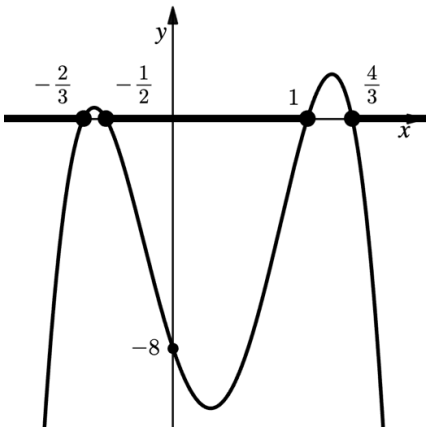
$$(4x^2-1)(1-9x^2)<0$$
$$\Rightarrow (2x-1)(2x+1)(1-3x)(1+3x)<0$$



$$\{x : x \in \mathbb{R}, x < -\frac{1}{2}\} \cup \{x : x \in \mathbb{R}, -\frac{1}{3} < x < \frac{1}{3}\} \cup \{x : x \in \mathbb{R}, x > \frac{1}{2}\}$$

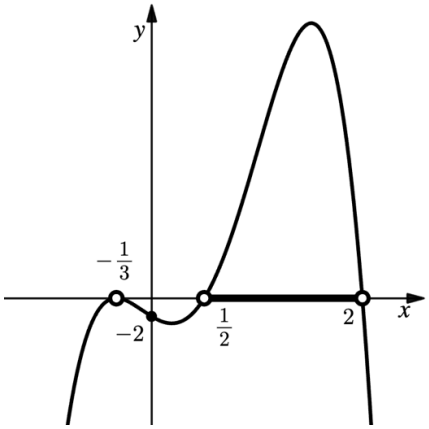
11 d)

$$(1+x-2x^2)(9x^2-6x-8) \leq 0$$
$$\Rightarrow (1-x)(2x+1)(3x+2)(3x-4) \leq 0$$



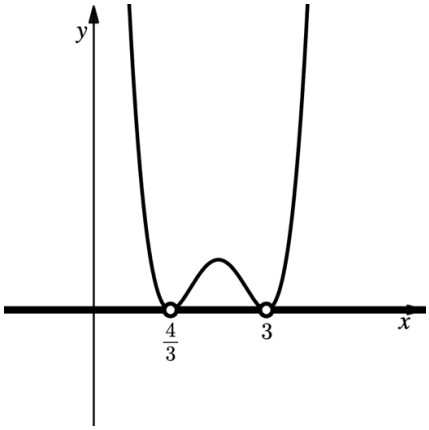
$$\{x : x \in \mathbb{R}, x \leq -\frac{2}{3}\} \cup \{x : x \in \mathbb{R}, -\frac{1}{2} \leq x \leq 1\} \cup \{x : x \in \mathbb{R}, x \geq \frac{4}{3}\}$$

11 e)



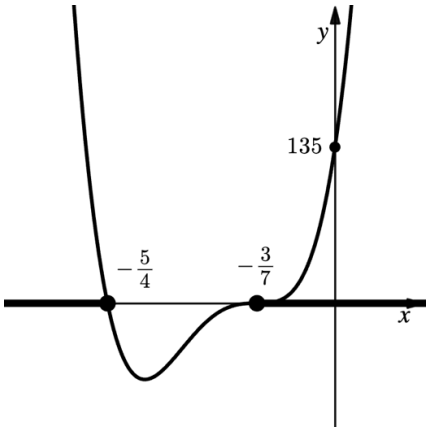
$$\{x : x \in \mathbb{R}, \frac{1}{2} < x < 2\}$$

11 f)



$$\{x : x \in \mathbb{R}, x < \frac{4}{3}\} \cup \{x : x \in \mathbb{R}, \frac{4}{3} < x < 3\} \cup \{x : x \in \mathbb{R}, x > 3\}$$

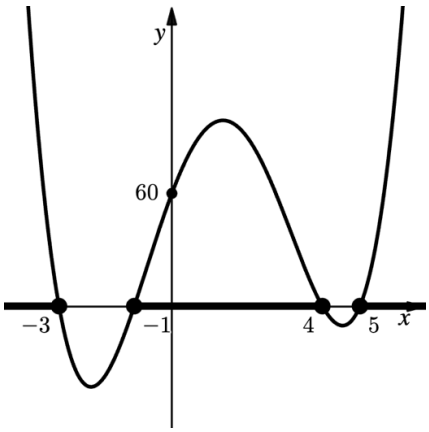
11 g)



$$\{x : x \in \mathbb{R}, x < -\frac{5}{4}\} \cup \{x : x \in \mathbb{R}, x > -\frac{3}{7}\}$$

12 a)

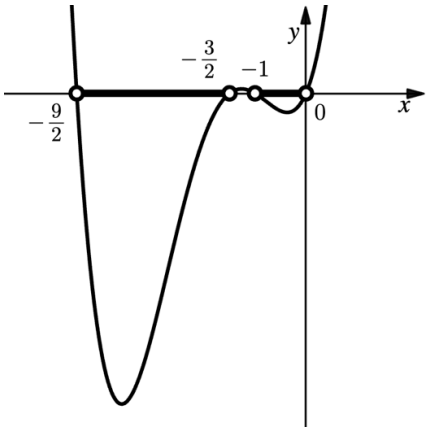
$$x^4 - 5x^3 - 13x^2 + 53x + 60 \geq 0$$
$$\Rightarrow (x+1)(x+3)(x-4)(x-5) \geq 0$$



$$x \in (-\infty, 3] \cup [-1, 4] \cup [5, \infty)$$

12 b)

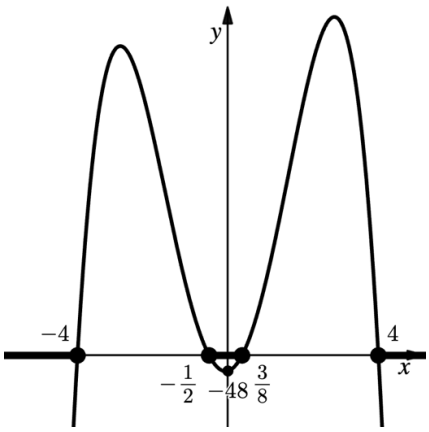
$$4x^4 + 28x^3 + 51x^2 + 27x < 0$$
$$\Rightarrow (2x+3)(2x+9)(x+1)x < 0$$



$$x \in (-\frac{9}{2}, -\frac{3}{2}) \cup (-1, 0)$$

12 c)

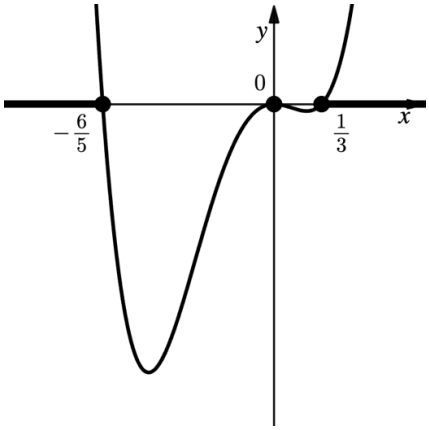
$$259x^2 + 32x - 48 - 2x^3 - 16x^4 \leq 0$$
$$\Rightarrow (3-8x)(2x+1)(x-4)(x+4) \leq 0$$



$$x \in (-\infty, -4] \cup [-\frac{1}{2}, \frac{3}{8}] \cup [4, \infty)$$

12 d)

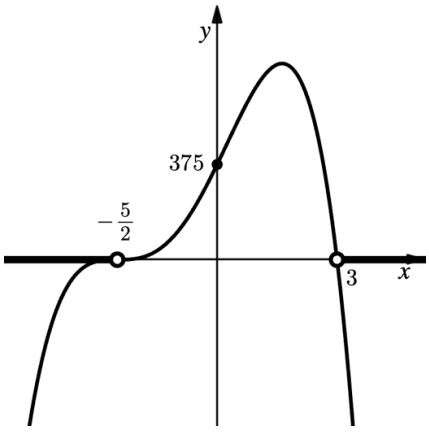
$$15x^4 + 13x^3 - 6x^2 \geq 0$$
$$\Rightarrow (5x+6)(3x-1)x^2 \geq 0$$



$$x \in (-\infty, -\frac{6}{5}] \cup \{0\} \cup [\frac{1}{3}, \infty) \text{ or } x \in (-\infty, -\frac{6}{5}] \cup [0,0] \cup [\frac{1}{3}, \infty)$$

12 e)

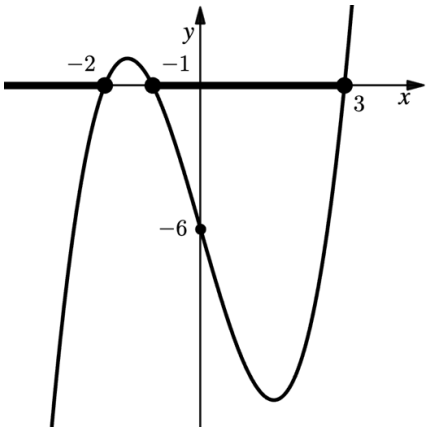
$$30x^2 + 325x + 375 - 36x^3 - 8x^4 < 0$$
$$\Rightarrow (2x+5)^3(3-x) < 0$$



$$x \in (-\infty, -\frac{5}{2}) \cup (3, \infty)$$

13 a)

$$(x+2)(x-3)(x+1)(x-5) \geq (x+2)^2(x-3)(x+1)$$
$$\Rightarrow (x+2)(x-3)(x+1)(x-5) - (x+2)^2(x-3)(x+1) \geq 0$$
$$\Rightarrow (x+2)(x-3)(x+1)((x-5) - (x+2)) \geq 0$$
$$\Rightarrow (x+2)(x-3)(x+1) \times -7 \geq 0$$
$$\Rightarrow (x+2)(x-3)(x+1) \leq 0$$

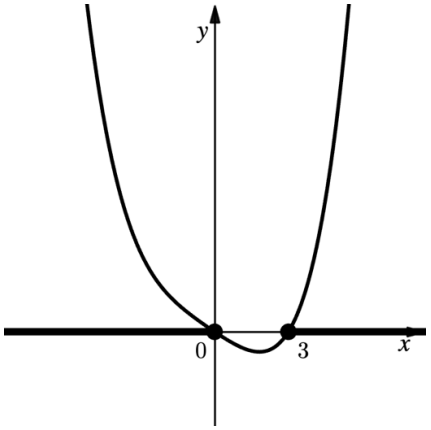


$$\{x : x \in \mathbb{R}, x \leq -2\} \cup \{x : x \in \mathbb{R}, -1 \leq x \leq 3\}$$

13 b)

$$
\begin{aligned}
&(5+x)(x+6)(3-x)x \leq (x-3)^2x^2 \\
\Rightarrow\ &(5+x)(x+6)(3-x)x - (x-3)^2x^2 \leq 0 \\
\Rightarrow\ &-(5+x)(x+6)(x-3)x - (x-3)^2x^2 \leq 0 \\
\Rightarrow\ &(5+x)(x+6)(x-3)x + (x-3)^2x^2 \geq 0 \\
\Rightarrow\ &x(x-3)((5+x)(x+6) + (x-3)x) \geq 0 \\
\Rightarrow\ &x(x-3)((x^2+11x+30) + (x^2-3x)) \geq 0 \\
\Rightarrow\ &x(x-3)(2x^2+8x+30) \geq 0 \\
\Rightarrow\ &x(x-3)(x^2+4x+15) \geq 0
\end{aligned}
$$

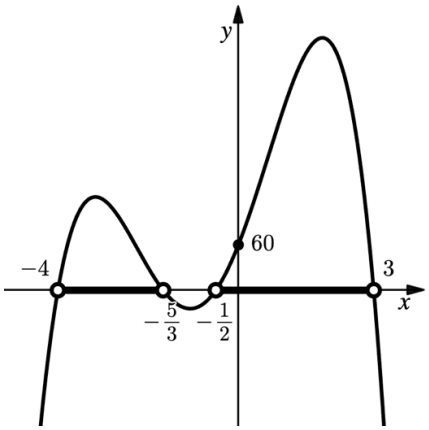
For $x^2+4x+15=0$, $b^2-4ac = 4^2 - 4\times 1\times 15 = -44 < 0$, so there are no real roots



$$\{x : x \in \mathbb{R}, x \leq 0\} \cup \{x : x \in \mathbb{R}, x \geq 3\}$$

13 c)

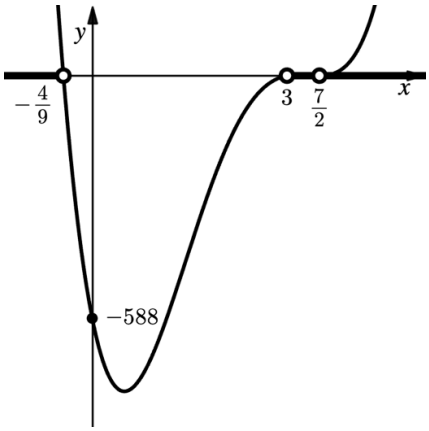
$$(3-x)(2x+1)(4+x)^2 > (x-3)(2x+1)^2(4+x)$$
$$\Rightarrow (3-x)(2x+1)(4+x)^2 - (x-3)(2x+1)^2(4+x) > 0$$
$$\Rightarrow (3-x)(2x+1)(4+x)^2 + (3-x)(2x+1)^2(4+x) > 0$$
$$\Rightarrow (3-x)(2x+1)(4+x)((4+x)+(2x+1)) > 0$$
$$\Rightarrow (3-x)(2x+1)(4+x)(3x+5) > 0$$



$$\{x : x \in \mathbb{R}, -4 < x < -\frac{5}{3}\} \cup \{x : x \in \mathbb{R}, -\frac{1}{2} < x < 3\}$$

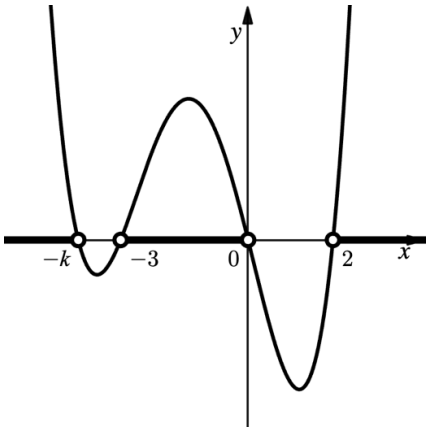
13 d)

$$(7-2x)^3(4+9x) < (2x-7)^2(4+9x)$$
$$\Rightarrow (7-2x)^3(4+9x) - (7-2x)^2(4+9x) < 0$$
$$\Rightarrow (7-2x)^2(4+9x)((7-2x)-1) < 0$$
$$\Rightarrow (7-2x)^2(4+9x)(6-2x) < 0$$
$$\Rightarrow (2x-7)^2(4+9x)(x-3) > 0$$



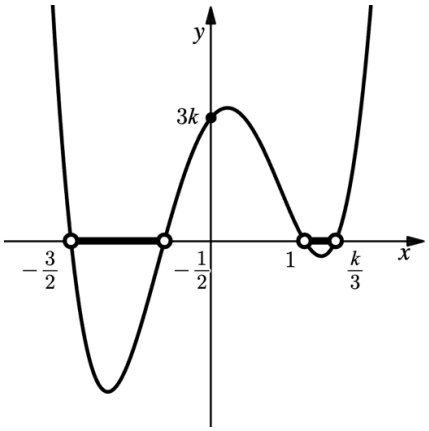
$$\{x : x \in \mathbb{R}, x < -\frac{4}{9}\} \cup \{x : x \in \mathbb{R}, 3 < x < \frac{7}{2}\} \cup \{x : x \in \mathbb{R}, x > \frac{7}{2}\}$$

14 a)



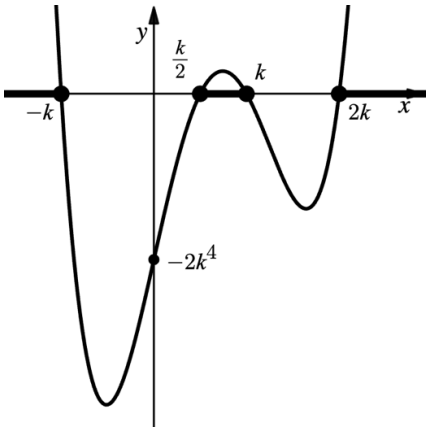
$$\{x : x \in \mathbb{R}, x < -k\} \cup \{x : x \in \mathbb{R}, -3 < x < 0\} \cup \{x : x \in \mathbb{R}, x > 2\}$$

14 b)



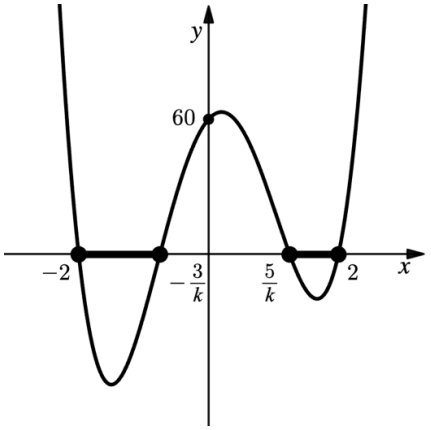
$$\{x : x \in \mathbb{R}, -\frac{3}{2} < x < -\frac{1}{2}\} \cup \{x : x \in \mathbb{R}, 1 < x < \frac{k}{3}\}$$

14 c)



$$\{x : x \in \mathbb{R}, x \leq -k\} \cup \{x : x \in \mathbb{R}, \frac{k}{2} \leq x \leq k\} \cup \{x : x \in \mathbb{R}, x \geq 2k\}$$

14 d)



$$\{x : x \in \mathbb{R}, -2 \leq x \leq -\frac{3}{k}\} \cup \{x : x \in \mathbb{R}, \frac{5}{k} \leq x \leq 2\}$$

15)
Critical values are $x = -\frac{3}{2}$, $x = -\frac{2}{5}$, $x = r$ and $x = \frac{3}{4}$

$$\therefore (2x+3)(5x+2)(x-r)(4x-3) < 0$$
$$\Rightarrow (5x+2)(x-r)(2x+3)(4x-3) < 0$$
$$\Rightarrow (5x^2 - 5rx + 2x - 2r)(8x^2 + 6x - 9) < 0$$
$$\Rightarrow (5x^2 + (2-5r)x - 2r)(8x^2 + 6x - 9) < 0$$
$$p = 5$$
$$2 - 5r = -4 \Rightarrow r = \frac{6}{5}$$
$$q = -2\left(\frac{6}{5}\right) = -\frac{12}{5}$$

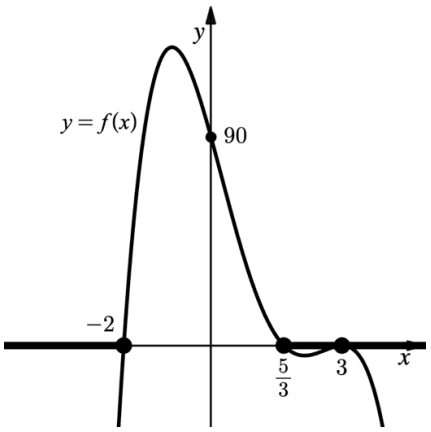
So $p = 5$, $q = -\frac{12}{5}$ and $r = \frac{6}{5}$

16)
Critical values are $x = -\frac{5}{2}$, $x = -\frac{3}{2}$, $x = r$ and $x = 4$

$$\begin{aligned}
&\therefore (2x+5)(2x+3)(x-r)(x-4) < 0 \\
&\Rightarrow (4x^2+6x+10x+15)(x^2-4x-rx+4r) < 0 \\
&\Rightarrow (4x^2+16x+15)(x^2-(4+r)x+4r) < 0 \\
&\Rightarrow 4x^4-(16+4r)x^3+16rx^2+16x^3-(64+16r)x^2 \\
&\qquad +64rx+15x^2-(60+15r)x+60r < 0 \\
&\Rightarrow 4x^4-4rx^3-49x^2+(49r-60)x+60r < 0 \\
&p = 4 \\
&-4r = -12 \Rightarrow r = 3 \\
&49(3)-60 = 87 \text{ as required} \\
&q = 60(3) = 180
\end{aligned}$$

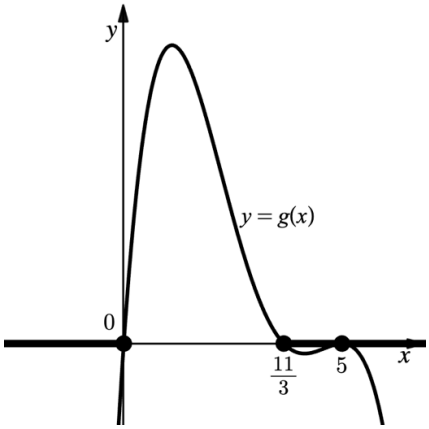
So $p = 4$, $q = 180$ and $r = 3$

17 a)



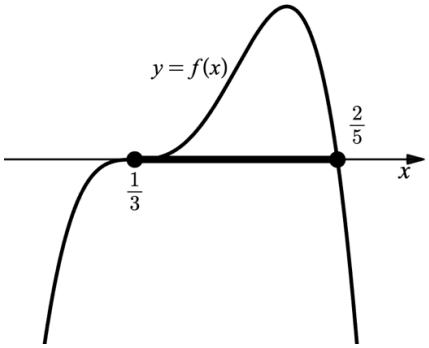
$$x \in (-\infty, -2] \cup [\frac{5}{3}, \infty)$$

17 b)



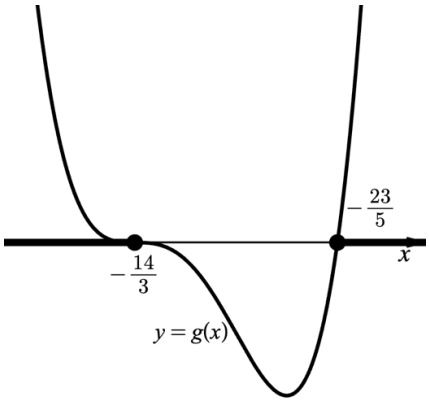
$$x \in (-\infty, 0] \cup [\frac{11}{3}, \infty)$$

18 a)



$$x \in [\frac{1}{3}, \frac{2}{5}]$$

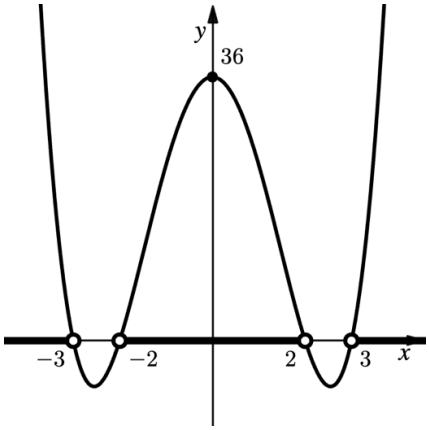
18 b)



$$x \in (-\infty, -\frac{14}{3}] \cup [-\frac{23}{5}, \infty)$$

19 a)

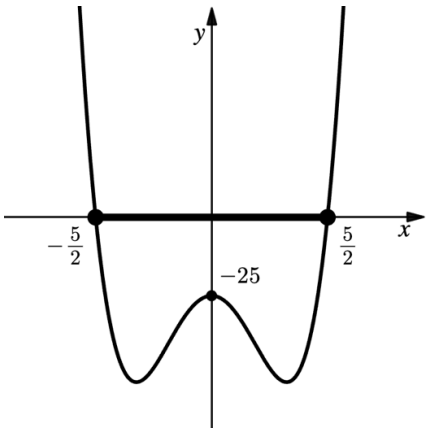
$$x^4 - 13x^2 + 36 > 0$$
$$\Rightarrow (x^2 - 4)(x^2 - 9) > 0$$
$$\Rightarrow (x - 2)(x + 2)(x - 3)(x + 3) > 0$$



$$\{x : x \in \mathbb{R}, x < -3\} \cup \{x : x \in \mathbb{R}, -2 < x < 2\} \cup \{x : x \in \mathbb{R}, x > 3\}$$

19 b)

$$4x^4 - 21x^2 - 25 \leq 0$$
$$\Rightarrow (x^2+1)(4x^2-25) \leq 0$$
$$\Rightarrow (x^2+1)(2x-5)(2x+5) \leq 0$$



$$\{x : x \in \mathbb{R}, -\frac{5}{2} < x < \frac{5}{2}\}$$

20 a)

$$e^{3x} - 12e^{2x} + 41e^x - 30 \leq 0$$
$$\Rightarrow (e^x - 1)(e^x - 5)(e^x - 6) \leq 0$$
$$e^x - 1 = 0 \Rightarrow e^x = 1 \Rightarrow x = 0$$
$$e^x - 5 = 0 \Rightarrow e^x = 5 \Rightarrow x = \ln 5$$
$$e^x - 6 = 0 \Rightarrow e^x = 6 \Rightarrow x = \ln 6$$

Critical values at $x = 0$, $x = \ln 5$ and $x = \ln 6$

$x < 0$	$0 < x < \ln 5$	$\ln 5 < x < \ln 6$	$x > \ln 6$
$-$	$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, x \leq 0\} \cup \{x : x \in \mathbb{R}, \ln 5 \leq x \leq \ln 6\}$$

20 b)

$$e^{6x} - 6e^{4x} - e^{2x} + 30 > 0$$
$$\Rightarrow (e^{2x} + 2)(e^{2x} - 3)(e^{2x} - 5) > 0$$

$e^{2x} + 2 = 0$ has no real solutions as $e^{2x} > 0$ for all $x \in \mathbb{R}$

$$e^{2x} - 3 = 0 \Rightarrow e^{2x} = 3 \Rightarrow 2x = \ln 3 \Rightarrow x = \frac{1}{2}\ln 3$$
$$e^{2x} - 5 = 0 \Rightarrow e^{2x} = 5 \Rightarrow 2x = \ln 5 \Rightarrow x = \frac{1}{2}\ln 5$$

Critical values at $x = \frac{1}{2}\ln 3$ and $x = \frac{1}{2}\ln 5$

$x < \frac{1}{2}\ln 3$	$\frac{1}{2}\ln 3 < x < \frac{1}{2}\ln 5$	$x > \frac{1}{2}\ln 5$
$+$	$-$	$+$

$$\left\{x : x \in \mathbb{R}, x < \frac{1}{2}\ln 3\right\} \cup \left\{x : x \in \mathbb{R}, x > \frac{1}{2}\ln 5\right\}$$

20 c)

$$e^{4x} - 4e^{3x} - 25e^{2x} + 88e^{x} - 60 \geq 0$$
$$\Rightarrow (e^{x} - 1)(e^{x} - 2)(e^{x} + 5)(e^{x} - 6) \geq 0$$
$$e^{x} - 1 = 0 \Rightarrow e^{x} = 1 \Rightarrow x = 0$$
$$e^{x} - 2 = 0 \Rightarrow e^{x} = 2 \Rightarrow x = \ln 2$$

$e^{x} + 5 = 0$ has no real solutions as $e^{x} > 0$ for all $x \in \mathbb{R}$

$$e^{x} - 6 = 0 \Rightarrow e^{x} = 6 \Rightarrow x = \ln 6$$

Critical values at $x = 0$, $x = \ln 2$ and $x = \ln 6$

$x < 0$	$0 < x < \ln 2$	$\ln 2 < x < \ln 6$	$x > \ln 6$
$-$	$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, 0 \leq x \leq \ln 2\} \cup \{x : x \in \mathbb{R}, x \geq \ln 6\}$$

20 d)

$$4e^{8x} - 8e^{6x} - 59e^{4x} + 63e^{2x} + 180 < 0$$
$$\Rightarrow (2e^{2x} + 3)(2e^{2x} - 5)(e^{2x} + 3)(e^{2x} - 4) < 0$$

$2e^{2x} + 3 = 0$ has no real solutions as $e^{2x} > 0$ for all $x \in \mathbb{R}$

$$2e^{2x} - 5 = 0 \Rightarrow e^{2x} = \frac{5}{2} \Rightarrow 2x = \ln\frac{5}{2} \Rightarrow x = \frac{1}{2}\ln\frac{5}{2}$$

$e^{2x} + 3 = 0$ has no real solutions as $e^{2x} > 0$ for all $x \in \mathbb{R}$

$$e^{2x} - 4 = 0 \Rightarrow e^{2x} = 4 \Rightarrow 2x = \ln 4 \Rightarrow x = \frac{1}{2}\ln 4 = \ln 2$$

Critical values at $x = \frac{1}{2}\ln\frac{5}{2}$ and $x = \ln 2$

$x < \frac{1}{2}\ln\frac{5}{2}$	$\frac{1}{2}\ln\frac{5}{2} < x < \ln 2$	$x > \ln 2$
$+$	$-$	$+$

$$\left\{x : x \in \mathbb{R}, \frac{1}{2}\ln\frac{5}{2} < x < \ln 2\right\}$$

21)

$$2^{12x} - 10 \times 2^{9x} + 160 \times 2^{3x} - 256 \leq 0$$
$$\Rightarrow (2^{3x} + 4)(2^{3x} - 2)(2^{3x} - 4)(2^{3x} - 8) \leq 0$$
$$2^{3x} + 4 \neq 0$$
$$2^{3x} - 2 = 0 \Rightarrow 2^{3x} = 2 \Rightarrow 3x = 1 \Rightarrow x = \frac{1}{3}$$
$$2^{3x} - 4 = 0 \Rightarrow 2^{3x} = 4 \Rightarrow 3x = 2 \Rightarrow x = \frac{2}{3}$$
$$2^{3x} - 8 = 0 \Rightarrow 2^{3x} = 8 \Rightarrow 3x = 3 \Rightarrow x = 1$$

Critical values at $x = \frac{1}{3}$, $x = \frac{2}{3}$ and $x = 1$

$x < \frac{1}{3}$	$\frac{1}{3} < x < \frac{2}{3}$	$\frac{2}{3} < x < 1$	$x > 1$
$-$	$+$	$-$	$+$

$$x \in (-\infty, \frac{1}{3}] \cup [\frac{2}{3}, 1]$$

22)

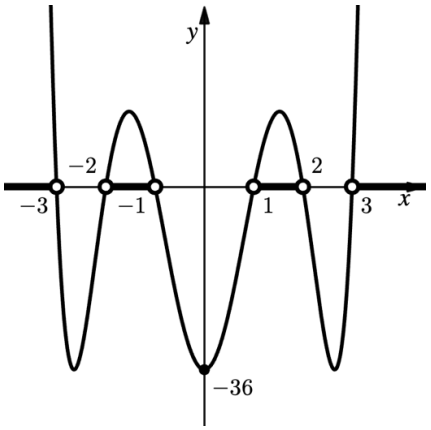
$$(2\sin x - 1)(\sin x + 3)(\sin x + k) > 0$$
$$2\sin x - 1 > 0 \Rightarrow \sin x > \frac{1}{2}$$

Has critical values when $\sin x = \frac{1}{2} \Rightarrow x = \left\{\frac{\pi}{6}, \frac{5\pi}{6}\right\}$
So $\sin x > \frac{1}{2}$ when $\frac{\pi}{6} < x < \frac{5\pi}{6}$
Meanwhile, $\sin x + 3 > 0$ for all $x \in \mathbb{R}$
So we need $\sin x + k > 0$ for all $x \in \mathbb{R}$
Minimum value of $\sin x$ is -1 so $-1 + k > 0 \Rightarrow k > 1$

23 a)

$$x^6 - 14x^4 + 49x^2 - 36 > 0$$
$$\Rightarrow (x^2 - 1)(x^2 - 4)(x^2 - 9) > 0$$
$$\Rightarrow (x - 1)(x + 1)(x - 2)(x + 2)(x - 3)(x + 3) > 0$$

$x < -3$	$-3 < x < -2$	$-2 < x < -1$	$-1 < x < 1$	$1 < x < 2$	$2 < x < 3$	$x > 3$
$+$	$-$	$+$	$-$	$+$	$-$	$+$



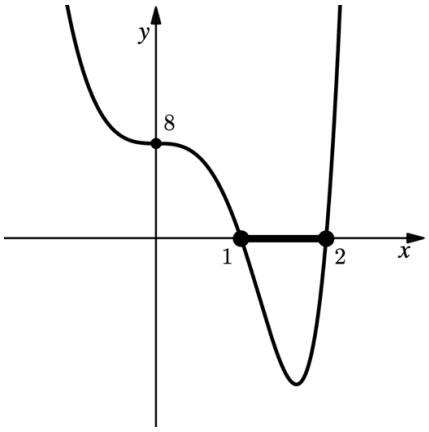
$$\{x : x \in \mathbb{R}, x < -3\} \cup \{x : x \in \mathbb{R}, -2 < x < -1\}$$
$$\cup \{x : x \in \mathbb{R}, 1 < x < 2\} \cup \{x : x \in \mathbb{R}, x > 3\}$$

23 b)

$$x^6 - 9x^3 + 8 \leq 0$$
$$\Rightarrow (x^3 - 1)(x^3 - 8) \leq 0$$

Critical values at $x = 1$ and $x = 2$

$x < 1$	$1 < x < 2$	$x > 2$
$+$	$-$	$+$



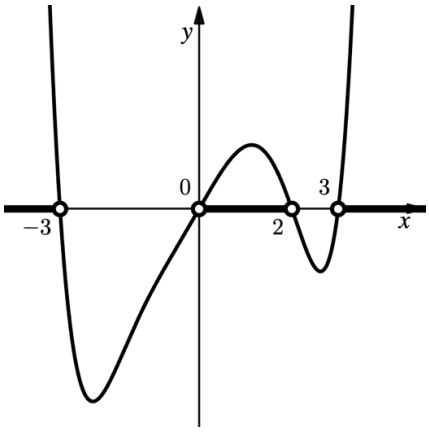
$$\{x : x \in \mathbb{R}, 1 \leq x \leq 2\}$$

23 c)

$$x^6 - 9x^4 - 8x^3 + 72x > 0$$
$$\Rightarrow (x^2 + 2x + 4)(x - 3)(x + 3)(x - 2)x > 0$$

$x^2 + 2x + 4 = 0$ has $b^2 - 4ac = 2^2 - 4 \times 1 \times 4 = -12 < 0$ so has no real roots.
$\therefore$ Critical values at $x = -3$, $x = 0$, $x = 2$ and $x = 3$

$x < -3$	$-3 < x < 0$	$0 < x < 2$	$2 < x < 3$	$x > 3$
$+$	$-$	$+$	$-$	$+$



$$\{x : x \in \mathbb{R}, x < -3\} \cup \{x : x \in \mathbb{R}, 0 < x < 2\} \cup \{x : x \in \mathbb{R}, x > 3\}$$